

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details <Do not mention any personally identifiable information like email ID, institute details, mobile phone number etc.>

Batch Number: 1 SmartStock Inventory Optimization for Retail Stores

Start date : 10-11-2025

Names:

1. Varsha Pamisetty
2. Rudrakshi Jaiswal
3. Jyotsana Sai
4. K. Deepak Reddy
5. Vijay

Internship Duration: 8 Weeks

1. Project Title

SmartStock – AI-Based Inventory Optimization System for Retail Stores

2. Project Objective

SmartStock is an AI-driven inventory optimization system designed to help retail stores efficiently manage stock levels, reduce overstock and stockouts, and improve overall operational efficiency. By analyzing historical sales data, demand patterns, seasonal trends, and supplier lead times, SmartStock predicts future product demand with high accuracy. The system provides intelligent stock replenishment recommendations and visual insights through interactive dashboards and reports. It also generates timely alerts for low stock, excess inventory, and fast-moving items. As new sales data is added, the AI model continuously learns and adapts to changing customer behavior. Supporting multiple store and product categories, SmartStock enables retailers to make data-driven inventory decisions, minimize losses, and maximize profitability.

3. Project description in detail

SmartStock is an intelligent inventory optimization solution developed to address common challenges faced by retail businesses such as overstocking, understocking, and inefficient inventory planning. The project leverages Artificial Intelligence (AI) and Machine Learning (ML) techniques to analyse historical sales data, identify demand trends, and accurately forecast future stock requirements.

The system collects and processes sales transactions, product details, supplier information, and inventory movement data. Using data preprocessing and classification techniques, products are grouped into categories such as fast-moving, slow-moving, and seasonal items. Predictive models are then applied to estimate future demand while considering factors like seasonal variations, promotional periods, and changing customer preferences.

A key feature of SmartStock is its interactive dashboard, which provides retailers with real-time insights through charts, graphs, and inventory reports. Store managers receive alerts for low stock levels, excess inventory, and reorder points, enabling timely restocking decisions. The system also helps in supplier planning by estimating optimal reorder quantities and lead times.

SmartStock supports multiple user roles, including store owners, managers, and inventory staff, making it suitable for single-store retailers as well as small retail chains. The AI model continuously improves its accuracy as more data is collected, adapting to evolving sales patterns and market trends.

Overall, SmartStock aims to transform traditional inventory management into a smart, automated, and data-driven process. By combining predictive analytics with a user-friendly interface, the project helps retailers reduce operational costs, improve product availability, and enhance customer satisfaction.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement gathering, problem analysis, and project scope definition	Retail inventory requirements finalised, project scope documented, system workflow planned
Week 2	Research on demand forecasting models and selection of development tools/tech stack	XGBoost model selected for sales forecasting, tech stack finalised (Django REST Framework, React Vite, MySQL, Python)
Week 3	Database design and setup for products, sales, and inventory data	MySQL database schema designed and tested for sales, inventory, and product tables
Week 4	Dataset collection, preprocessing, and feature engineering	Store sales time-series dataset (t4tiana/store-sales-time-series-forecasting) collected, cleaned, and transformed
Week 5	Model development: Training ML model for sales and demand forecasting	XGBoost model trained and evaluated using historical sales data
Week 6	Frontend development: Dashboard, graphs, and inventory views	React Vite dashboard developed with inventory status and demand forecast visualisations using Matplotlib
Week 7	System integration: Backend APIs, ML model, database, and frontend	Django REST APIs integrated with ML model and frontend, forecasting and stock alerts enabled

Week	Activities Planned	Activities Completed
Week 8	Testing, debugging, documentation, and final deployment	End-to-end testing completed, bugs fixed, documentation prepared, project deployed successfully

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project kickoff conducted to define SmartStock objectives, inventory optimization scope, system workflow, and individual responsibilities.	14 – 11 – 2025
Prototype / First Draft	Developed initial prototype including database schema, basic Django REST APIs, and sample sales forecasting output using XGBoost.	28 – 11 – 2025
Mid-Term Review	Mid-term review conducted to evaluate data preprocessing, ML model performance, backend–frontend integration progress, and improvements required.	12 – 12 – 2025
Final Submission	Submitted the completed SmartStock Inventory Optimisation system with trained XGBoost model, React dashboard, APIs, and documentation.	26 – 12 – 2025
Presentation	Prepared and delivered the final project presentation demonstrating inventory forecasting, dashboards, and system outcomes.	02 – 01 – 2026

5b. Project execution details

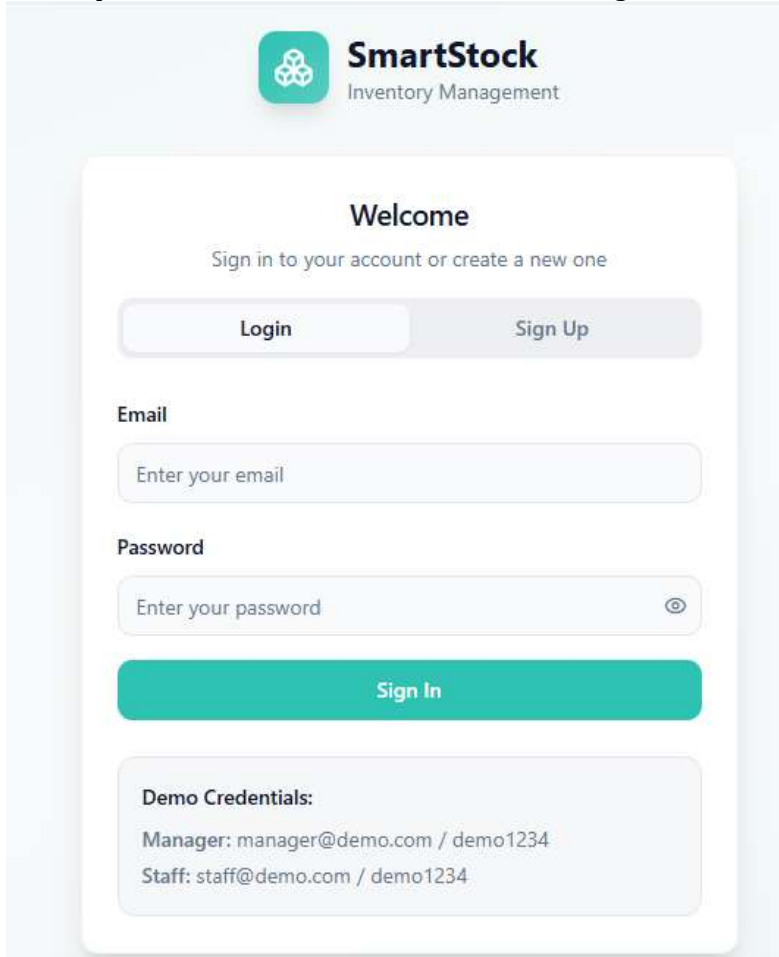
- **Requirement Analysis** – Analyzed retail inventory challenges, gathered user requirements, defined project scope, and finalized objectives for inventory optimization.
- **Technology Selection** – Selected Django REST Framework for backend APIs, React (Vite) for frontend, MySQL for database management, XGBoost for demand forecasting, and Matplotlib for data visualization.
- **System Design** – Designed database schema for products, sales, and inventory, created system workflow diagrams, and planned frontend UI structure.
- **Data Collection** – Collected store sales time-series dataset from *t4tiana/store-sales-time-series-forecasting* for training and testing.
- **Data Preprocessing** – Cleaned and transformed sales data, handled missing values, created time-based features, and structured data for ML model training.
- **Model Development** – Trained and evaluated the XGBoost model for sales and demand forecasting, tuned model parameters, and validated prediction accuracy.
- **Frontend Development** – Developed a responsive React dashboard to display inventory status, demand forecasts, and graphs using Matplotlib outputs.
- **Backend Development** – Built REST APIs using Django REST Framework, integrated the trained ML model, and connected backend services with the frontend and database.

- **Testing & Integration** – Performed unit testing, API testing, and end-to-end system integration; identified and fixed functional and performance issues.

6. Snapshots / Screenshots

Step 1: Open the SmartStock Application

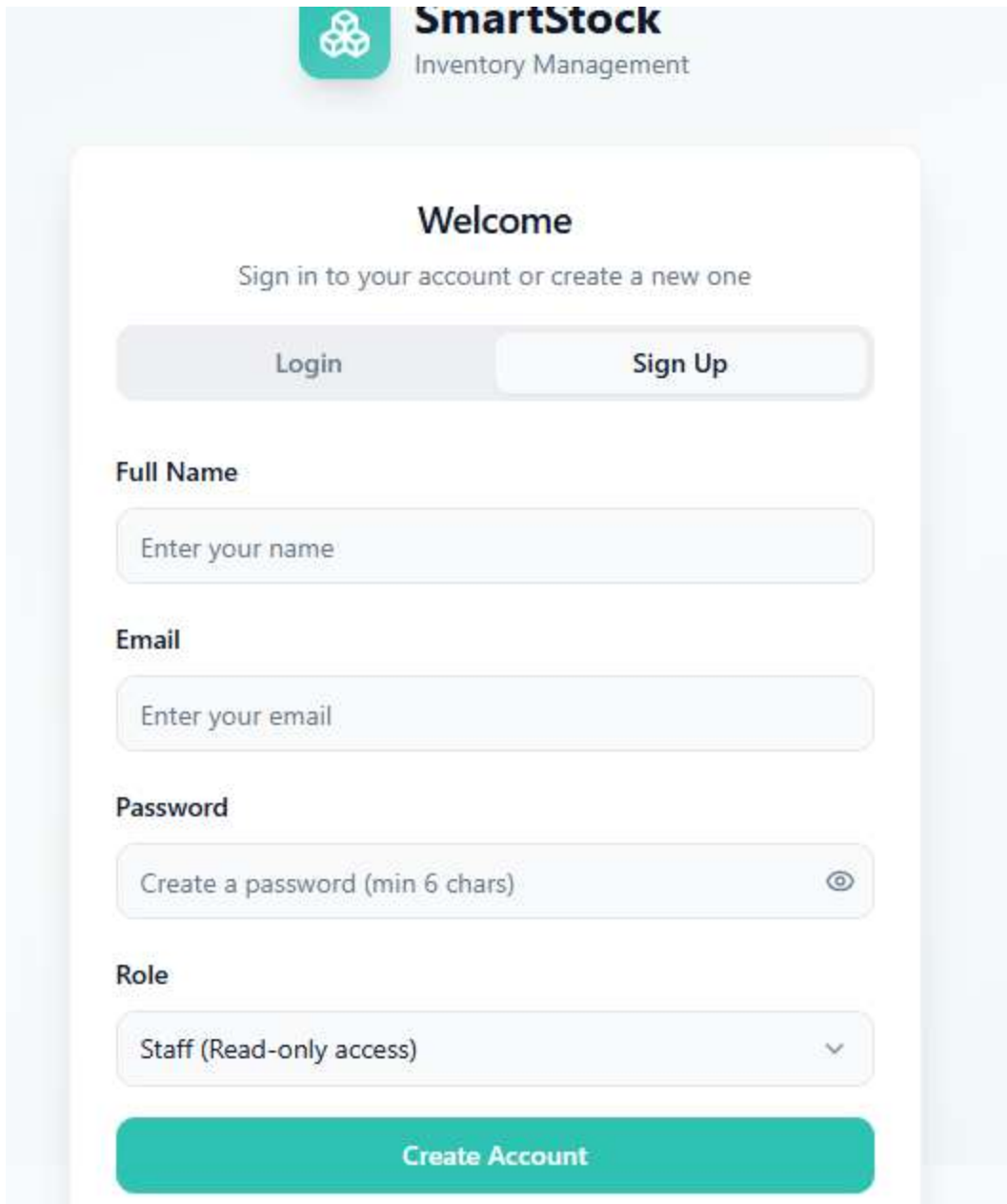
In this step, the user opens the **SmartStock Inventory Management System**. The landing screen provides access to various modules through the left-side navigation menu



The screenshot shows the login interface of the SmartStock Inventory Management System. At the top, there is a logo with a green cube icon and the text "SmartStock Inventory Management". Below the logo, the word "Welcome" is displayed, followed by the instruction "Sign in to your account or create a new one". There are two buttons: "Login" and "Sign Up". Below these buttons are input fields for "Email" and "Password". The "Email" field has a placeholder "Enter your email". The "Password" field has a placeholder "Enter your password" and a toggle icon. Below the input fields is a large green "Sign In" button. At the bottom, there is a section titled "Demo Credentials:" with the following text: "Manager: manager@demo.com / demo1234" and "Staff: staff@demo.com / demo1234".

Step 2: Login to the System

The user logs into the application using valid credentials. Role-based authentication ensures that only authorized users such as managers can access inventory functions.



The image shows a web application interface for 'SmartStock Inventory Management'. At the top left is a logo with three interlocking circles. To its right, the text 'SmartStock' is in a large, bold, black font, with 'Inventory Management' in a smaller, regular black font below it. The main content area is a white card with a light gray border. Inside the card, the word 'Welcome' is centered in a bold black font. Below it, the text 'Sign in to your account or create a new one' is centered in a smaller, regular black font. There are two buttons: 'Login' on the left and 'Sign Up' on the right, both in a light gray box with rounded corners. Below these buttons are four form fields, each with a label above it: 'Full Name' (with placeholder text 'Enter your name'), 'Email' (with placeholder text 'Enter your email'), 'Password' (with placeholder text 'Create a password (min 6 chars)' and an eye icon for toggling visibility), and 'Role' (a dropdown menu showing 'Staff (Read-only access)' with a downward arrow). At the bottom of the card is a large, teal-colored button with the text 'Create Account' in white.

SmartStock
Inventory Management

Welcome

Sign in to your account or create a new one

Login Sign Up

Full Name
Enter your name

Email
Enter your email

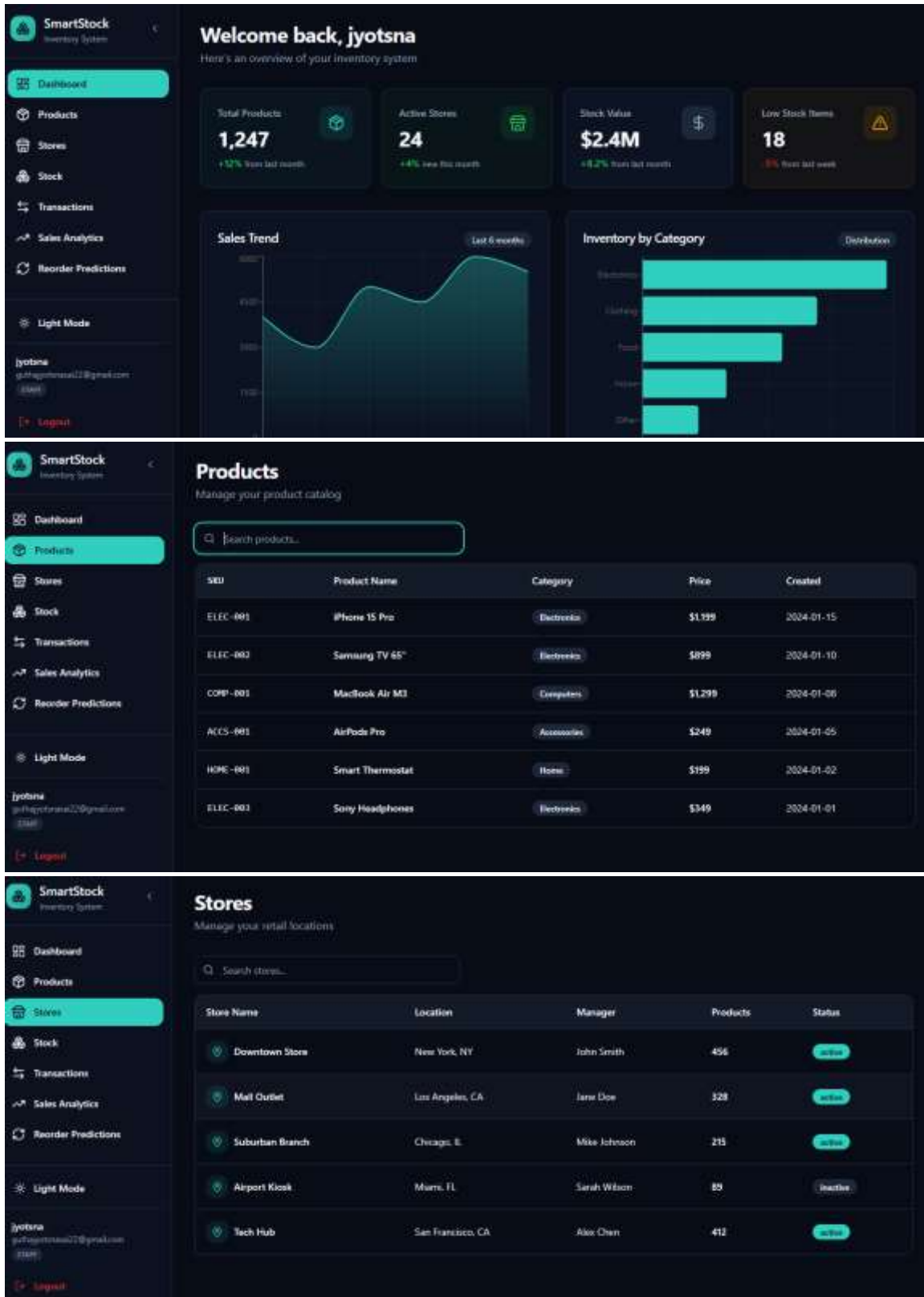
Password
Create a password (min 6 chars) 

Role
Staff (Read-only access) 

Create Account

Step 3: Dashboard Overview

After login, the dashboard displays a summary of total products, stores, stock status, and notifications. It provides a quick overview of inventory performance (staff account)



SmartStock Inventory System

Stock Management

Monitor and update inventory levels

Search products...

Product	Store	Quantity	Min. Qty	Status	
iPhone 15 Pro ELEC-001	Downtown Store	45	20	In Stock	2024-01-20
Samsung TV 65" ELEC-002	Downtown Store	8	10	Low Stock	2024-01-19
MacBook Air M3 COMP-001	Mall Outlet	23	15	In Stock	2024-01-18
AirPods Pro ACCS-001	Tech Hub	5	25	Low Stock	2024-01-17
Sony Headphones ELEC-003	Suburban Branch	67	20	In Stock	2024-01-16
Smart Thermostat HOME-001	Mall Outlet	3	10	Low Stock	2024-01-15

All Stores

- ✓ All Stores
- Downtown Store
- Mall Outlet
- Suburban Branch
- Tech Hub

Dashboard
Products
Stores
Stock
Transactions
Sales Analytics
Reorder Predictions

Light Mode

lyotona
lyotona22@gmail.com
(12345)

Logout

SmartStock Inventory System

Transactions

View and record inventory transactions

Search transactions...

Date	Type	Product	Quantity	Store	Recorded By
2024-01-20 14:30	Sale	iPhone 15 Pro ELEC-001	-5	Downtown Store	John Smith
2024-01-20 11:15	Restock	AirPods Pro ACCS-001	+100	Tech Hub	Jane Doe
2024-01-19 16:45	Sale	MacBook Air M3 COMP-001	-3	Mall Outlet	Mike Johnson
2024-01-19 10:00	Transfer	Samsung TV 65" ELEC-002	-10	Downtown Store	Sarah Wilson
2024-01-18 09:30	Adjustment	Smart Thermostat HOME-001	-2	Mall Outlet	Alice Chen
2024-01-17 15:20	Sale	Sony Headphones ELEC-003	-8	Suburban Branch	John Smith

Dashboard
Products
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Light Mode

lyotona
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(12345)

Logout

SmartStock Inventory System

Sales Analytics

Analyze sales trends and performance

Select Product

iPhone 15 Pro

✓ iPhone 15 Pro
Samsung TV 65"
MacBook Air M3
AirPods Pro
Sony Headphones

Total Sales (8 months): **1,235 units**

Monthly Average: **206 units**

Trend: **+5.8%**

Revenue Trend

Dashboard
Products
Stores
Stock
Transactions
Sales Analytics
Reorder Predictions

Light Mode

lyotona
lyotona22@gmail.com
(12345)

Logout

Step 4: Viewing Sales Analytics

This screen shows the **Sales Analytics page** where users analyze sales trends and performance for different products.



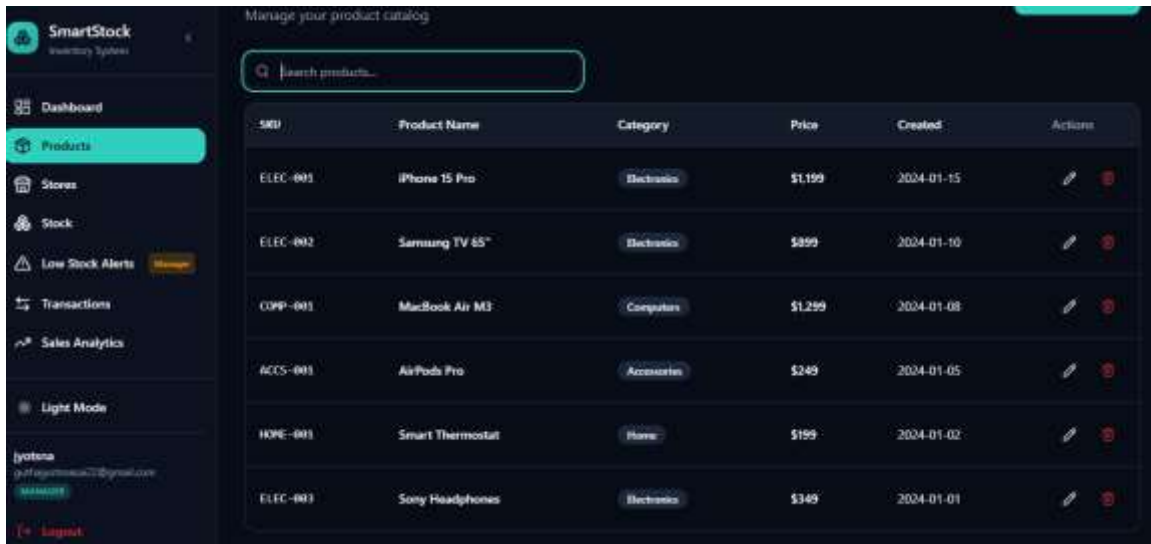
Step 5: Sales Trend Graph

This screen displays the **Sales Trend – iPhone 15 Pro** line graph, showing monthly sales patterns over six months.



Step 6: Adding a New Product

The user can add new inventory items by entering product details such as name, category, price, and quantity.



The screenshot displays the 'SmartStock' inventory system interface. On the left is a dark sidebar with navigation links: Dashboard, Products (highlighted in red), Stores, Stock, Low Stock Alerts (with a Manage button), Transactions, Sales Analytics, and Light Mode. The main area is titled 'Manage your product catalog' and features a search bar. Below the search bar is a table with columns: SKU, Product Name, Category, Price, Created, and Actions. The table lists six products: iPhone 15 Pro, Samsung TV 65", MacBook Air M3, AirPods Pro, Smart Thermostat, and Sony Headphones. Each product row includes edit and delete icons in the Actions column.

SKU	Product Name	Category	Price	Created	Actions
ELEC-001	iPhone 15 Pro	Electronics	\$1,199	2024-01-15	
ELEC-002	Samsung TV 65"	Electronics	\$899	2024-01-10	
COMP-001	MacBook Air M3	Computers	\$1,299	2024-01-08	
ACCS-001	AirPods Pro	Accessories	\$249	2024-01-05	
HOME-001	Smart Thermostat	Home	\$199	2024-01-02	
ELEC-003	Sony Headphones	Electronics	\$349	2024-01-01	

Step 7: Low Stock Alerts

The system generates alerts for products that reach minimum stock levels, helping avoid stockouts and lost sales.

The screenshot displays the 'SmartStock' inventory system's 'Low Stock Alerts' dashboard. The dashboard includes a sidebar with navigation links (Dashboard, Products, Stores, Stock, Low Stock Alerts, Transactions, Sales Analytics, Light Mode) and a main content area. The main area shows a 'Threshold' of 10 and a 'Send Email Alert' button. It features three summary cards: 'Critical Alerts' (2 items, 2 days until stockout), 'Warning Alerts' (2 items, 5-9 days until stockout), and 'Total Alerts' (4 items, below threshold of 10). Below these are three product rows: 'Samsung TV 65\"

Low Stock Alerts
Monitor and manage inventory alerts

Threshold: 10 [Send Email Alert](#)

Critical Alerts
2 items
2 days until stockout

Warning Alerts
2 items
5-9 days until stockout

Total Alerts
4 items
Below threshold of 10

Product	Current	Minimum	Suggested Order	Days left
Samsung TV 65" (ELEC-002) Chowtown Store	8	10	25	5
AirPods Pro (ACCS-001) Tech Hub	5	25	50	2
Smart Thermostat (HOME-001) Mall Outlet	3	10	20	1

Send Low Stock Alert Email
Send an email notification with all current low stock alerts (4 items)

Recipient Email *
manager@example.com

Recipient Name
John Smith

Email will include:

- 2 critical alerts
- 2 warning alerts
- 0 low priority alerts
- Suggested reorder quantities

[Cancel](#) [Send Email](#)

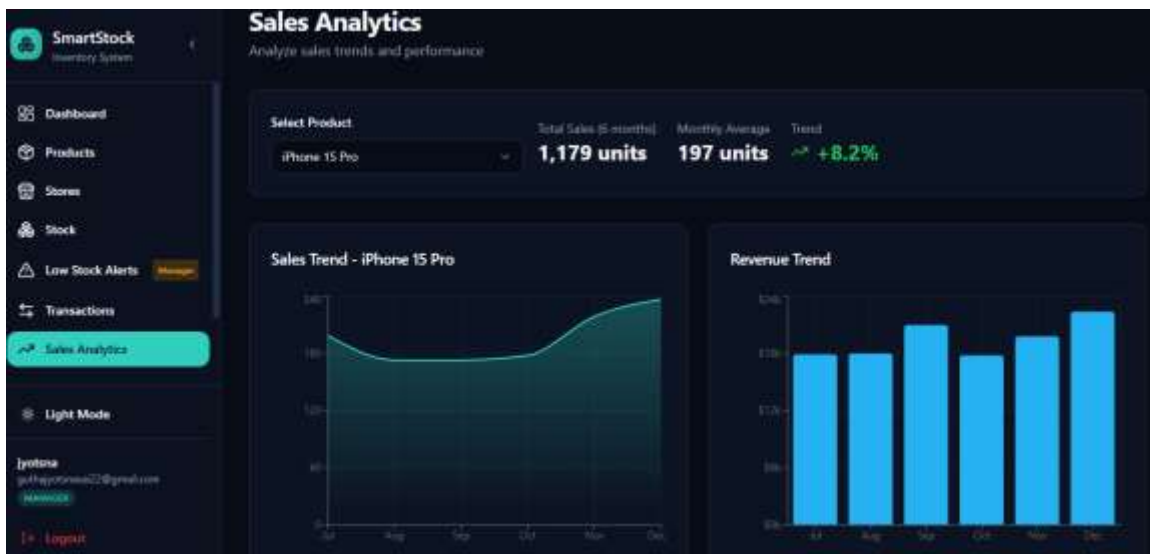
Step 8: Adding Stock Transactions

This screen shows stock-in and stock-out transactions being recorded for proper tracking of inventory levels.

Date	Type	Product	Quantity	Store	Recorded By
2024-01-20 14:30	Sale	iPhone 15 Pro (128G - 981)	-5	Downtown Store	John Smith
2024-01-20 11:15	Restock	AirPods Pro (2nd Gen - 981)	+100	Tech Hub	Jane Doe
2024-01-19 16:45	Sale	MacBook Air M3 (128G - 981)	-3	Mall Outlet	Mike Johnson
2024-01-19 10:00	Transfer	Samsung TV 65" (128G - 982)	-10	Downtown Store	Sarah Wilson
2024-01-18 09:30	Adjustment	Smart Thermostat (256G - 983)	-2	Mall Outlet	Alex Chen

Step 9: Revenue Trend Graph

This screenshot shows the **Revenue Trend bar chart**, representing monthly revenue comparison based on product sales.



7. Challenges Faced

- **Data Availability** – Accessing real-world retail inventory datasets with complete sales and demand details was challenging, so an open-source store sales time-series dataset was used.

- **Data Quality Issues** – The dataset contained missing values, irregular time intervals, and inconsistent sales records, requiring extensive data cleaning and preprocessing.
- **Demand Forecasting Accuracy** – Achieving accurate demand predictions was difficult due to seasonal variations, promotions, and sudden changes in customer buying behavior.
- **Model Selection** – Selecting the most suitable ML model for sales forecasting required experimentation before finalizing XGBoost for better performance and accuracy.
- **Integration Complexity** – Integrating the XGBoost ML model with Django REST APIs, MySQL database, and React frontend required careful handling of data flow and synchronization.
- **Performance Optimization** – Ensuring fast API responses and efficient ML predictions without slowing down the system was a key challenge.
- **Frontend Visualization** – Presenting inventory status and forecast results clearly using graphs required multiple iterations for better clarity and usability.
- **Backend–Frontend Communication** – Managing smooth data exchange between Django REST backend and React (Vite) frontend needed proper API design and testing.
- **Time Management** – Completing data preprocessing, model training, frontend development, backend integration, and testing within the 8-week internship period was demanding.
- **Testing & Validation** – Validating demand forecasts across different products and time periods required repeated testing and evaluation.

8. Learnings & Skills Acquired

- **Machine Learning Skills** – Gained practical experience in demand forecasting using the XGBoost algorithm on time-series sales data.
- **Data Preprocessing & Feature Engineering** – Learned how to clean sales data, handle missing values, and create time-based features for ML models.
- **Time-Series Analysis** – Developed an understanding of sales trends, seasonality, and demand patterns in retail inventory systems.
- **Backend Development** – Strengthened skills in building RESTful APIs using Django REST Framework and integrating ML models into backend services.
- **Frontend Development** – Enhanced React (Vite) skills by creating dashboards to display inventory data and forecast results.

- **Database Management** – Improved knowledge of designing and managing relational databases using MySQL for inventory and sales data.
- **Data Visualization** – Learned to generate and interpret graphs using Matplotlib for sales trends and demand forecasts.
- **Problem-Solving** – Improved ability to handle real-world issues such as data inconsistency, integration bugs, and performance bottlenecks.
- **Testing & Debugging** – Gained experience in API testing, unit testing, and end-to-end system validation.
- **Project Management** – Learned to plan tasks, meet deadlines, and manage a complete AI-based project within a fixed timeline.

9. Testimonials from team

Team Member 1:

"This project helped me understand how machine learning models like XGBoost can be applied to real-world retail sales forecasting and inventory optimization."

Team Member 2:

"I gained hands-on experience in data preprocessing, backend API development, and integrating ML models with a web application."

Team Member 3:

"SmartStock improved my understanding of full-stack development and taught me how to connect frontend, backend, and machine learning components effectively."

Team Member 4:

"The project enhanced my skills in data analysis, visualization, and problem-solving while working on a real-world inventory management use case."

Team Member 5:

"The project improved my knowledge in NLP, data preprocessing, and visualization techniques, and also boosted my confidence in handling real-world AI applications."

10. Conclusion

The SmartStock Inventory Optimization project successfully delivered an AI-based system for forecasting product demand and improving inventory management in retail stores. By combining machine learning (XGBoost), time-series sales analysis, Django REST APIs, React-based dashboards, and MySQL database management, the system enables data-driven inventory decisions. The project enhanced technical knowledge, analytical thinking, and practical development skills while providing a scalable and efficient solution to common retail inventory challenges.

11. Acknowledgements

I would like to express my sincere gratitude to **Infosys** for providing me the opportunity to work on the **SmartStock Inventory Optimization** project and gain valuable hands-on experience in AI, machine learning, and full-stack development.

I am especially thankful to my mentor Shakthi Gopalakrishnan, for her continuous guidance, support, and valuable feedback throughout the internship. Her mentorship played a crucial role in overcoming technical challenges and improving my understanding of real-world applications.

I would also like to thank my team members for their cooperation, support, and collaborative efforts in successfully completing this project.

Finally, I extend my appreciation to everyone who contributed directly or indirectly to my learning and the successful completion of this internship.